

Button

DETECT

INPUT

What does it do?

A button lets someone tell your program to do something the moment they press.

Inside are two metal contacts. Pressing pushes them together so electricity can flow, and letting go springs them apart again.

Your projects can use it to start something, stop something, or count how many times it has been pressed.

Uses



Pedestrian crossing

DETECT
Button and
traffic phase



DECIDE
Is it safe to
cross?



DO
Change the
lights



Do not press this

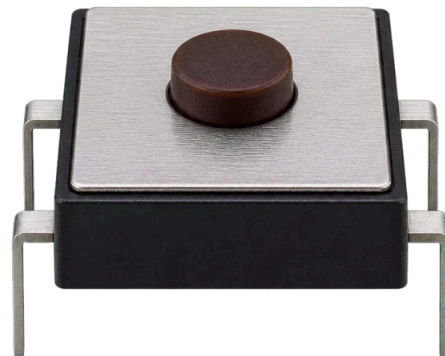
DETECT
Button is
pressed



DECIDE
Is the trap
armed?



DO
Play a silly
sound



Cap

Press here

Legs

Four, joined in
pairs

Gap

Straddle the
channel

Code

● Raspberry Pi

```
from gpiozero import Button

button = Button(24)

pressed = button.is_pressed
```

● Raspberry Pi Pico

```
from picozero import Button

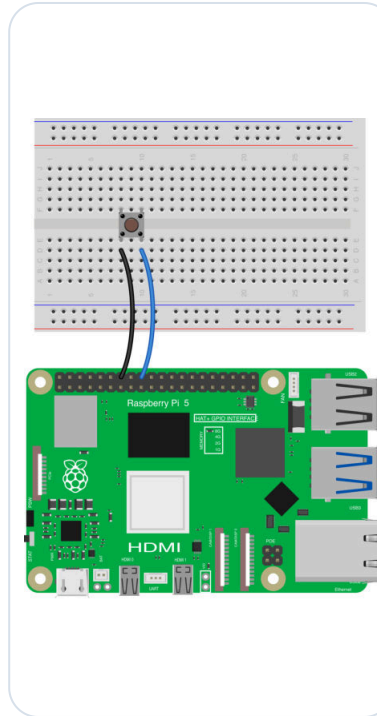
button = Button(16)

pressed = button.is_pressed
```

`pressed` is True or False. No resistor is needed — both libraries switch on the pin's own pull-up. Also `button.when_pressed` and `button.when_released`.

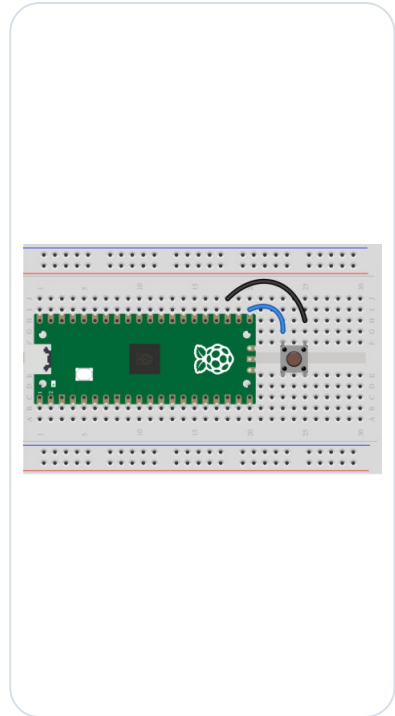
Wiring

RASPBERRY PI 5



GPIO 24 → button → GND

PICO



GP16 → button → GND

Capacitive touch

DETECT

INPUT

What does it do?

A capacitive touch sensor notices when a finger comes close to its metal pad.

It detects the tiny change in electric charge caused by your body, so no moving parts are needed.

Use it anywhere a project needs a hidden, wipe-clean or feather-light button.

Uses



Touch lamp

DETECT
Finger touches the pad

DECIDE
Is the lamp already on?

DO
Toggle the light



Banana piano

DETECT
A banana is touched

DECIDE
Which note should play?

DO
Play the note



Touch pad
The large metal spiral

SIG
Digital output

VCC and GND
3.3 V power

Code

● Raspberry Pi

```
from gpiozero import DigitalInputDevice

touch = DigitalInputDevice(16, pull_up=False)

touched = touch.is_active
```

● Raspberry Pi Pico

```
from picozero import TouchSensor

touch = TouchSensor(22)

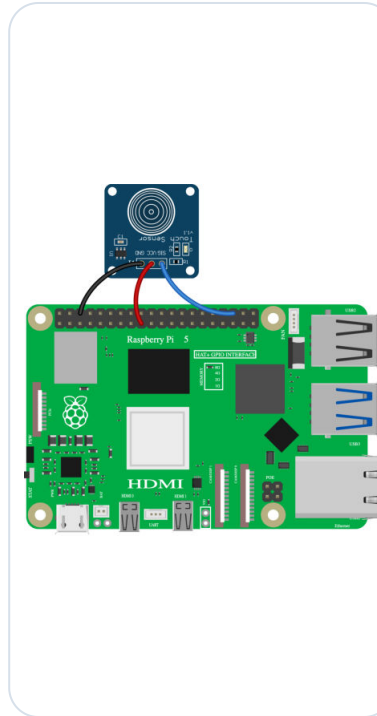
touched = touch.is_touched
```

The TTP223 output goes high while touched.

touched is True or False.

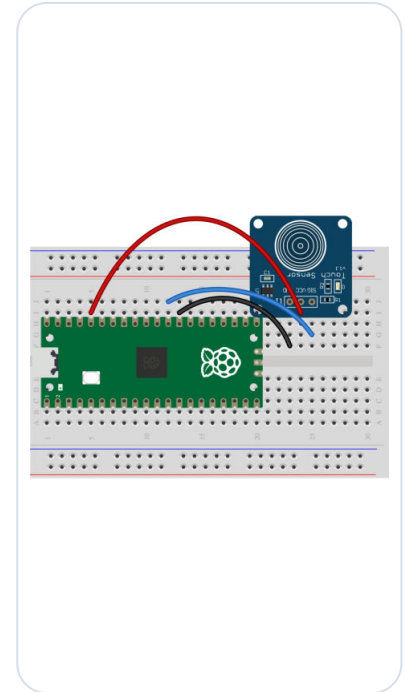
Wiring

RASPBERRY PI 5



GND → GND, **3V3** → VCC, **GPIO 16** → SIG

PICO



GND → GND, **3V3 OUT** → VCC, **GP22** → SIG

LDR

DETECT

LIGHT

What does it do?

An LDR is a sensor that measures how bright or dark it is.

LDR stands for light-dependent resistor. Light shining on it lets electricity through more easily, so your code can tell light from dark.

Your projects can use it to react when a lamp comes on, when night falls, or when a shadow passes over.

Uses



Street light

DETECT
How dark it is



DECIDE
Is it darker
than dusk?



DO
Switch the
lamp on



Biscuit tin alarm

DETECT
Light inside
the tin



DECIDE
Is the lid
open?



DO
Sound the
alarm



Window
Light lands
here

Track
Zigzag sensor

Legs
Either way
round

Code

● Raspberry Pi

```
from gpiozero import LightSensor

ldr = LightSensor(18)

light = ldr.value
```

● Raspberry Pi Pico

```
from picozero import AnalogInputDevice

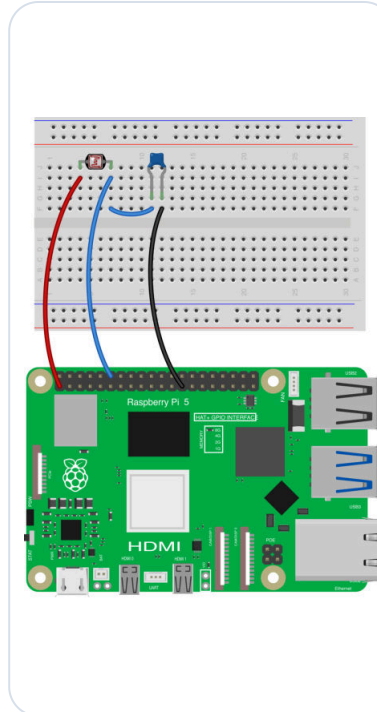
ldr = AnalogInputDevice(26)

light = ldr.value
```

`light` runs from 0 in the dark to 1 in bright light. A Pi also gives you `ldr.light_detected` and `ldr.when_light`; a Pico gives you `ldr.voltage`.

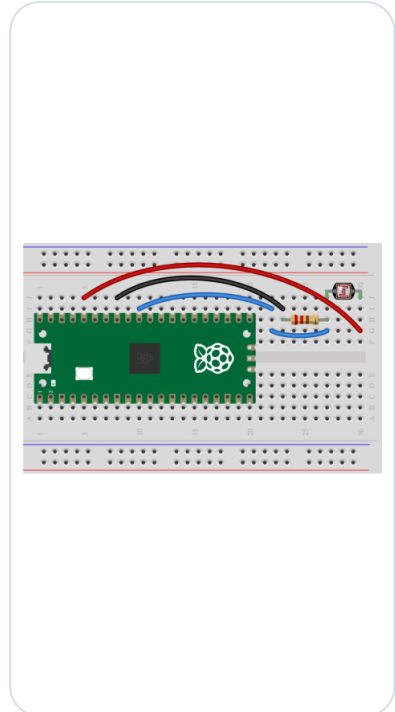
Wiring

RASPBERRY PI 5



3V3 → LDR → GPIO 18 → 1 μ F → GND

PICO



3V3 → LDR → GP26 → 10 k Ω → GND

Line sensor

DETECT

INPUT

What does it do?

A line sensor tells light and dark surfaces apart underneath a moving project.

An infrared LED shines at the floor and a detector measures how much light bounces back.

Adjust the small potentiometer until its digital output changes reliably at the line edge.

Uses



Warehouse guide-line robot

DETECT

Read guide-line position



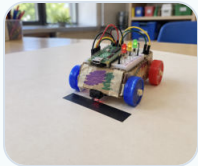
DECIDE

Is the vehicle drifting?



DO

Correct the steering



Cardboard lap-counting car

DETECT

Cross dark finish strip



DECIDE

Is this a new lap?



DO

Increase the lap count



Infrared pair
Faces the floor

Adjuster
Sets the threshold

S
Digital signal

Code

● Raspberry Pi

```
from gpiozero import LineSensor  
  
line = LineSensor(16)  
  
on_line = line.is_active
```

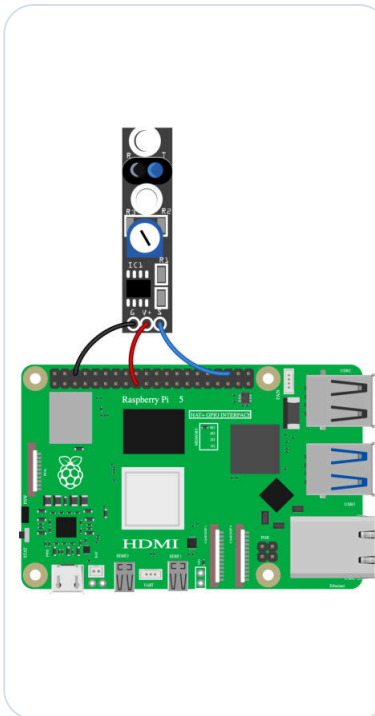
● Raspberry Pi Pico

```
from picozero import DigitalInputDevice  
  
line = DigitalInputDevice(22, pull_up=False)  
  
on_line = line.is_active
```

Turn the blue adjuster until the module's output changes cleanly at the edge of the line.

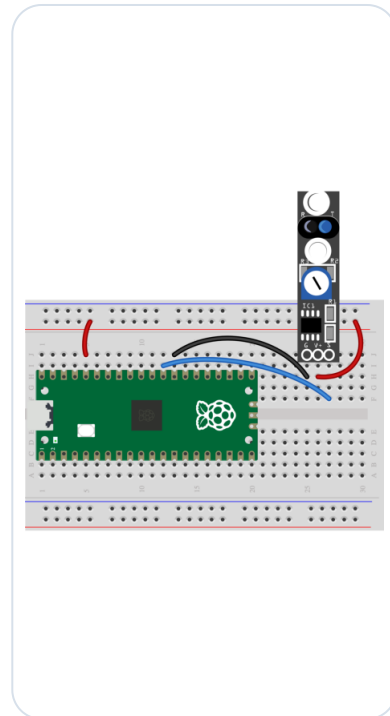
Wiring

RASPBERRY PI 5



GND → GND, **3V3** → V+, **GPIO 16** → S

PICO



GND → GND, **3V3 OUT** → V+, **GP22** → S

Distance sensor

DETECT

DISTANCE

What does it do?

A distance sensor measures how far away something is without touching it.

It works like a bat. One side sends out a click too high for you to hear, the other side listens for the echo, and the time it takes gives the distance.

Your projects can use it to react as something comes closer, or to measure a gap that keeps changing.

Uses



Parking sensor

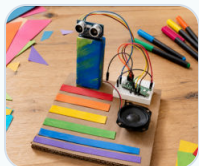
DETECT

Gap to the wall

DECIDE
Is it closer
than 30 cm?

DO

Beep faster



Air harp

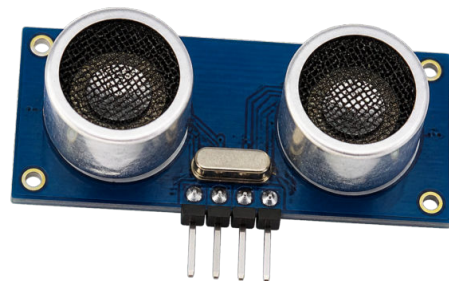
DETECT

How far your
hand is

DECIDE
Which note is
that?

DO

Play the note



VCC

Power,
3V3

Trig

Sends a
click

Echo

Times the
return

GND

Ground

Code

● Raspberry Pi

```
from gpiozero import DistanceSensor

sensor = DistanceSensor(
    echo=18, trigger=17
)

distance = sensor.distance
```

● Raspberry Pi Pico

```
from picozero import DistanceSensor

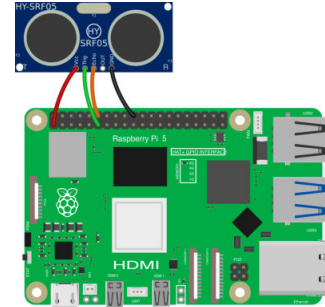
sensor = DistanceSensor(
    echo=15, trigger=13
)

distance = sensor.distance
```

`distance` is in metres and stops at 1. Pass `max_distance=2` to see further, or read `sensor.value` for 0 to 1. Power it from **3V3**, not 5V.

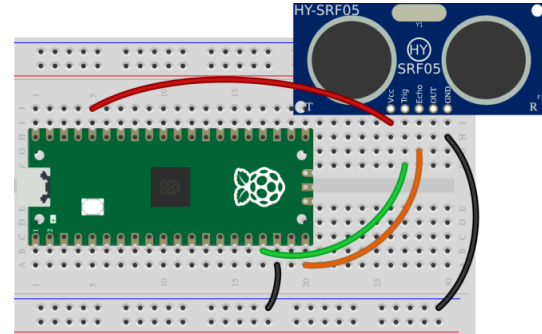
Wiring

RASPBERRY PI 5



Trig **GPIO 17** · Echo **GPIO 18**

PICO



Trig **GP13** · Echo **GP15**

PIR

DETECT

MOTION

What does it do?

A PIR spots movement nearby, such as somebody walking past.

PIR stands for passive infrared. It watches the warmth that bodies give off and reports when that warmth moves, so it notices people and animals rather than furniture.

Your projects can use it to wake something up the moment somebody arrives, and go quiet again when they leave.

Uses

Burglar alarm

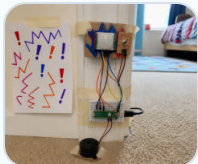


DETECT
Movement in
the room

DECIDE
Should the
room be
empty?

DO
Raise the
alarm

Sibling alarm



DETECT
Movement at
the door

DECIDE
Is privacy
mode armed?

DO
Sound the
warning



Dome
Wide-
angle lens

VCC
Power, 5V

OUT
High on
movement

GND
Ground

Code

● Raspberry Pi

```
from gpiozero import MotionSensor  
  
pir = MotionSensor(4)  
  
moved = pir.motion_detected
```

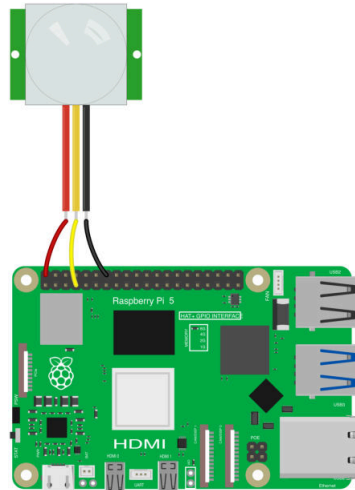
● Raspberry Pi Pico

```
from picozero import MotionSensor  
  
pir = MotionSensor(16)  
  
moved = pir.motion_detected
```

`moved` is True or False. Also `pir.when_motion` and `pir.when_no_motion` to run a function the moment it changes. Most boards carry two dials that set how sensitive it is and how long OUT stays high.

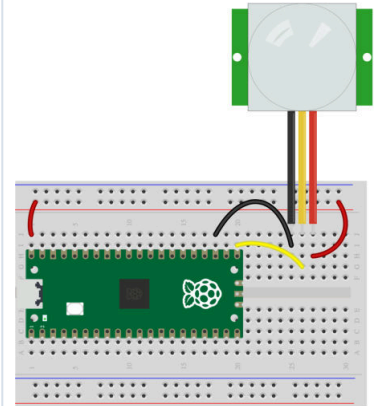
Wiring

RASPERRY PI 5



5V · OUT GPIO 4 · GND

PICO



VBUS · OUT GP16 · GND

Potentiometer

DETECT

INPUT

What does it do?

A potentiometer lets your code read how far a knob has been turned.

This small version is called a trimpot. Turning the blue knob moves a contact called the wiper, changing the voltage your board reads.

Use it to choose a speed, a brightness or a sound. Turn gently and stop at the ends. The marking 103 means 10 kΩ.

Uses



Security-light adjustment

DETECT
Read
sensitivity
setting



DECIDE
Is movement
strong
enough?



DO
Switch the
light on



Alien voice tuner

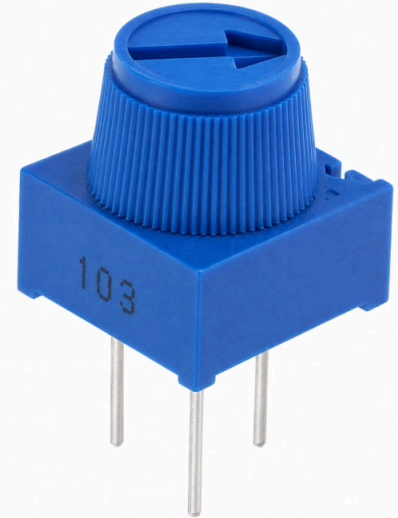
DETECT
Knob position



DECIDE
Should it
rumble or
squeak?



DO
Rumble or
squeak



Knob
Turn to adjust

Outer pair
Pins 1 and 3

Wiper
Pin 2, offset

Code

● Raspberry Pi

```
from gpiozero import MCP3008

pot = MCP3008(0, select_pin=22)

setting = pot.value
```

● Raspberry Pi Pico

```
from picozero import Pot

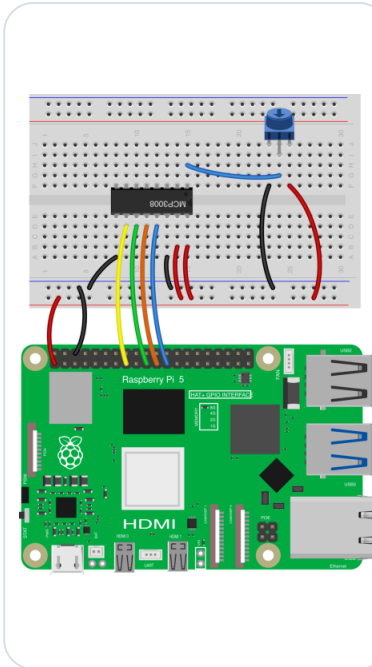
pot = Pot(26)

setting = pot.value
```

`setting` runs from 0 to 1. Read `pot.value` again after turning; `pot.voltage` gives volts. The Pi needs the MCP3008 converter shown. Power the trimpot and converter from **3V3**.

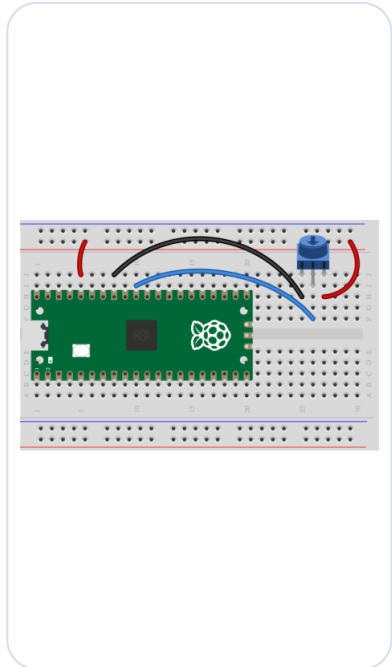
Wiring

PI 5 + MCP3008



Wiper → **CH0**. CLK / DIN / DOUT / CS → **GPIO 11 / 10 / 9 / 22**. Chip notch faces right.

PICO



Pin 1 → **GND** · 2 (wiper) → **GP26** · 3 → **3V3**. Each pin has its own breadboard row.

Temperature sensor

DETECT

INPUT

ANALOGUE

What does it do?

A TMP36 turns temperature into a changing voltage that a program can measure.

Its output rises by 10 millivolts for every degree Celsius, with 0.5 V representing 0 °C.

The Pico reads that voltage directly; the Raspberry Pi uses an MCP3008 analogue-to-digital converter.

Uses



Refrigerator cooling

DETECT
Measure
fridge
temperature

DECIDE
Is it too
warm?

DO
Increase the
cooling

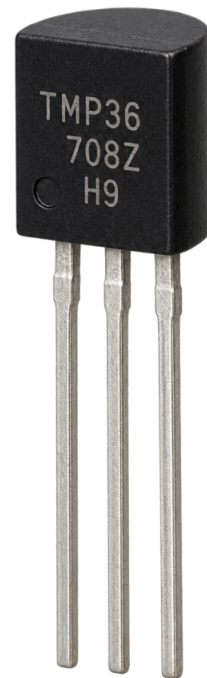


Soup drama meter

DETECT
Measure bowl
surface

DECIDE
Which drama
level fits?

DO
Move the
drama pointer



**Flat
face**

Use it to
identify
the pins

+Vs
Left pin

Vout
Centre pin

GND
Right pin

Code

● Raspberry Pi

```
from gpiozero import MCP3008

sensor = MCP3008(channel=0)

temperature = (sensor.voltage - 0.5) * 100
```

● Raspberry Pi Pico

```
from picozero import TemperatureSensor

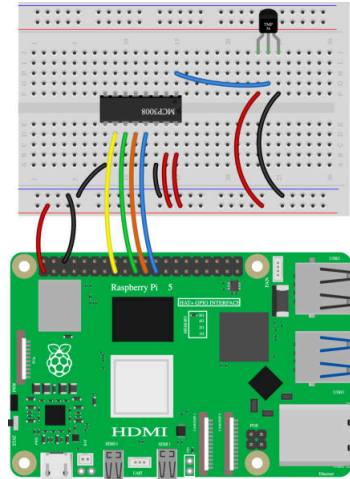
convert = lambda voltage: (voltage - 0.5) * 100
sensor = TemperatureSensor(26, conversion=convert)

temperature = sensor.temp
```

Read the TMP36 with its flat face towards you; the pins are +Vs, Vout, GND from left to right.

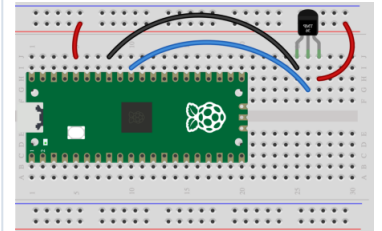
Wiring

RASPBERRY PI 5



TMP36 **Vout** → MCP3008 **CHO**; ADC SPI uses GPIO **22, 10, 9, 11**

PICO



TMP36 **+Vs** → 3V3 OUT, **Vout** → GP26, **GND** → GND

Tilt switch

DETECT

INPUT

What does it do?

A tilt switch changes state when it tips past a particular angle.

A tiny metal ball rolls inside and touches two contacts, completing the circuit.

It can detect that an object has been picked up, tipped over or opened.

Uses



Iron safety shut-off

DETECT
Iron tips
sideways



DECIDE
Has the delay
elapsed?



DO
Switch the
heater off



Grumpy pencil case

DETECT
Case is tipped



DECIDE
Has it stayed
tipped?



DO
Make a
grumpy noise



Metal can
Contains the
rolling ball

Legs
No polarity

Angle
Changes the
contact

Code

● Raspberry Pi

```
from gpiozero import Button  
  
tilt = Button(23)  
  
tilted = tilt.is_pressed
```

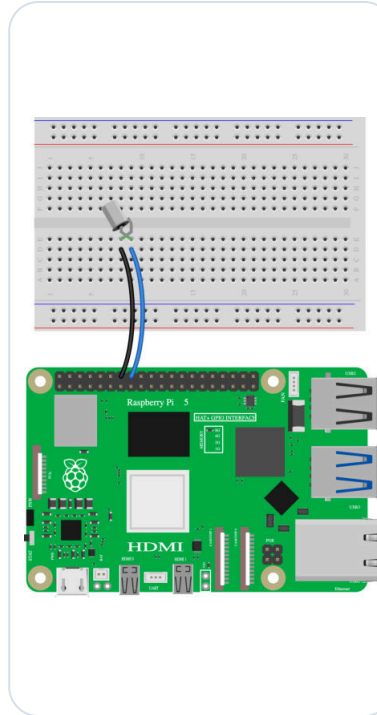
● Raspberry Pi Pico

```
from picozero import Switch  
  
tilt = Switch(16)  
  
tilted = tilt.is_closed
```

The reading may flicker while the ball rolls.
Check it more than once before treating it as a definite tilt.

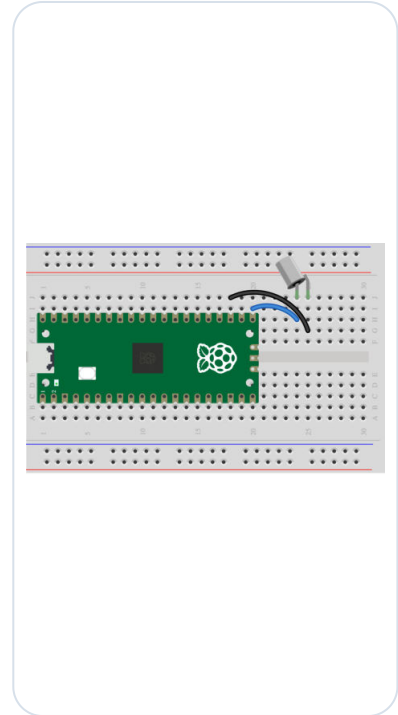
Wiring

RASPBERRY PI 5



GPIO 23 → tilt switch → GND

PICO



GP16 → tilt switch → GND

Toggle switch

DETECT

INPUT

What does it do?

A toggle switch stays in the position you move it to, unlike a push button.

Its centre COM leg connects to one outside leg in each position.

It is useful when a project needs a lasting on/off choice or two modes.

Uses



Auto/manual selector

DETECT

Read selector position



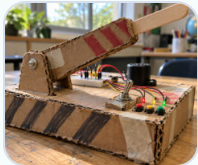
DECIDE

Is automatic mode selected?



DO

Run the selected mode



Serious business lever

DETECT

Lever and check buttons



DECIDE

Are all checks complete?



DO

Start unnecessary sirens



Lever

Stays where it is moved

COM

Centre pin

L1 and L2

The two choices

Code

● Raspberry Pi

```
from gpiozero import Button  
  
mode = Button(23)  
  
mode_two = mode.is_pressed
```

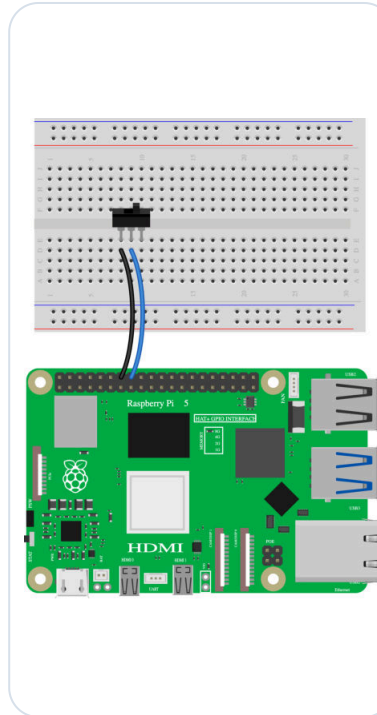
● Raspberry Pi Pico

```
from picozero import Switch  
  
mode = Switch(16)  
  
mode_two = mode.is_closed
```

Connect COM and one outside pin. The unused outside pin can be left unconnected.

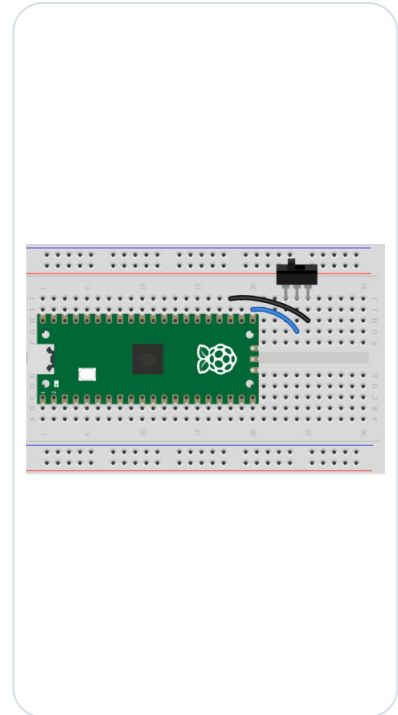
Wiring

RASPBERRY PI 5



GPIO 23 → COM; one outside pin → **GND**

PICO



GP16 → COM; one outside pin → **GND**

Pi Camera

DETECT

IMAGES

VIDEO

What does it do?

A camera turns light into digital images. Connect a Raspberry Pi Camera Module to a Raspberry Pi computer with a flat ribbon cable.

Use Python to take a photograph or record a video. Your program can also examine images and decide what to do next.

Uses



Video doorbell

DETECT

Camera sees movement.

DECIDE

Is a person there?

DO

Save a video clip.



Monster photo booth

DETECT

Button starts countdown.

DECIDE

Has time run out?

DO

Take a silly photo.



Lens

Focuses light

Sensor

Records the image

Ribbon

Connects to the Pi

Code

● Take photo

```
from picamera2 import Picamera2

camera = Picamera2()
camera.start_and_capture_file(
    "photo.jpg", delay=2,
    show_preview=False)
camera.close()
```

● Take video

```
from picamera2 import Picamera2

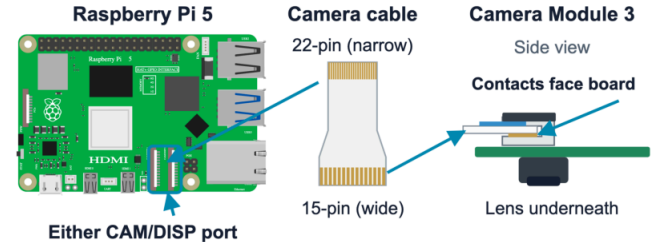
camera = Picamera2()
camera.start_and_record_video(
    "video.mp4", duration=5)
camera.close()
```

Run each example separately on **Raspberry Pi OS** with **Picamera2** installed. Files save in the current folder. The video is five seconds long, without sound.

Wiring

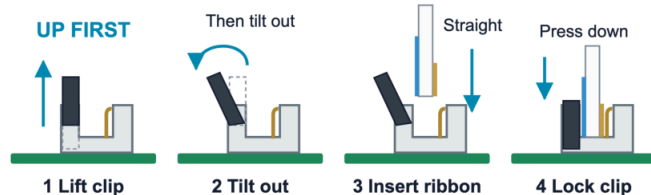
RASPBERRY PI 5 + CAMERA MODULE 3

POWER OFF • Unplug USB-C before fitting



Lift UP first → tilt → insert → lock

Pi 5 connector shown from the side



Blue backing faces the flap; metal contacts face away.

At the camera, contacts face towards the circuit board.

Seat both edges evenly. Do not force or crease the ribbon.

Use a **Standard-Mini camera cable** (22-pin to 15-pin). The wide cable supplied with the camera fits a Pi 4, not a Pi 5.

LED

DO

LIGHT

What does it do?

An LED is a tiny light that your code can switch on, switch off, or blink.

LED stands for light-emitting diode. It glows when electricity flows the right way through it, so the long leg and the short leg matter.

Your projects can use LEDs to show a warning, celebrate something, or tell people what is happening.

Uses



Car brake light

DETECT

Brake control
signal



DECIDE

Is braking
requested?



DO

Light the red
LEDs



One-eyed monster box

DETECT

Box lid is
opened



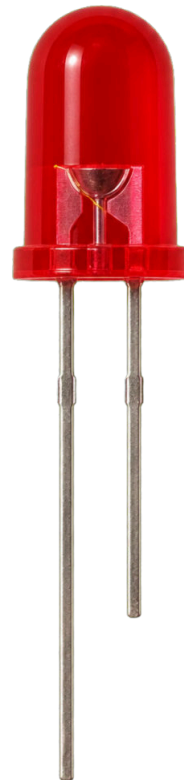
DECIDE

Is guard
mode armed?



DO

Glow the
monster eye



Anode

Long leg, to
GPIO

Cathode

Short leg, to
GND

Lens

Glow here

Code

● Raspberry Pi

```
from gpiozero import LED  
  
led = LED(17)  
  
led.on()
```

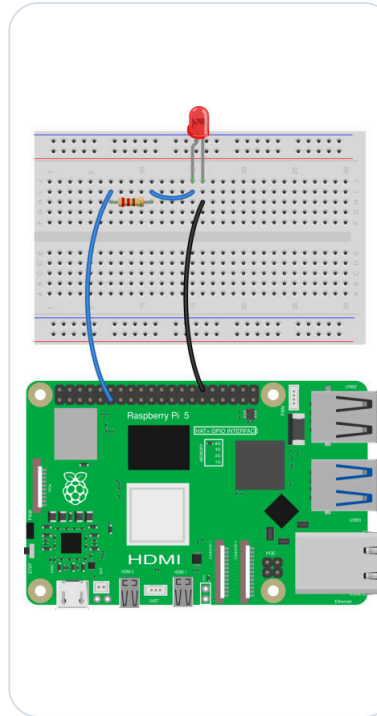
● Raspberry Pi Pico

```
from picozero import LED  
  
led = LED(16)  
  
led.on()
```

Also `led.off()`, `led.toggle()` and `led.blink()`. Give `blink()` an `on_time` and an `off_time` to set the flash speed.

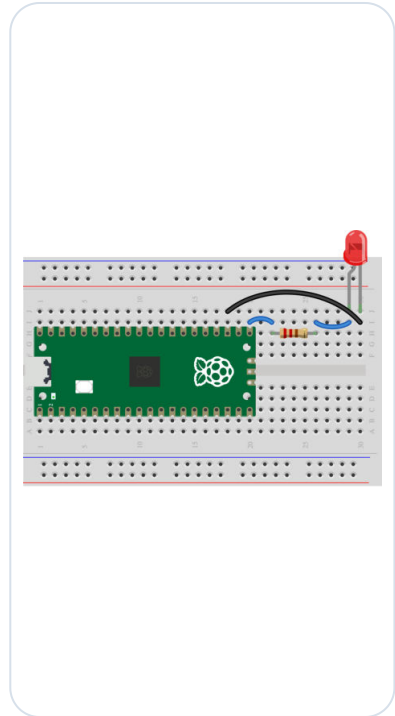
Wiring

RASPBERRY PI 5



GPIO 17 → 220 Ω → LED → GND

PICO



GP16 → 220 Ω → LED → GND

RGB LED

DO

OUTPUT

LIGHT

What does it do?

An RGB LED puts red, green and blue lights inside one clear package.

Varying the brightness of the three colours mixes millions of possible colours.

A common-cathode LED has one shared negative leg and needs one resistor for each colour.

Uses



Wi-Fi router status

DETECT
Read connection status

DECIDE
Which status colour is needed?

DO
Mix the LED colour



Mood crystal

DETECT
Read the mood dial

DECIDE
What is the crystal's mood?

DO
Glow that colour



Clear lens
Contains three LEDs

Long leg
Common cathode

R
One 220 ohm resistor each

Code

● Raspberry Pi

```
from gpiozero import RGBLED
light = RGBLED(red=17, green=27, blue=22)
light.color = (1, 0, 1)
```

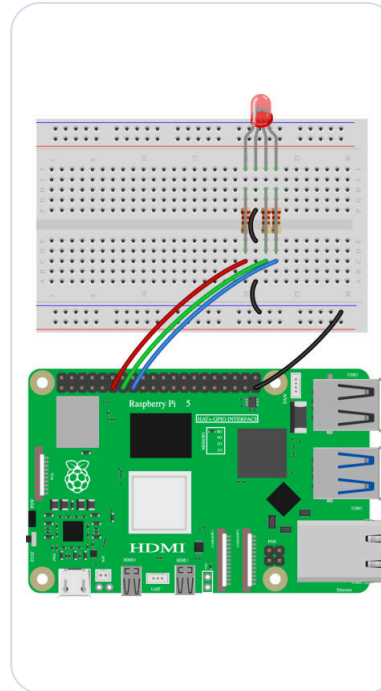
● Raspberry Pi Pico

```
from picozero import RGBLED
light = RGBLED(red=15, green=13, blue=10)
light.color = (255, 0, 255)
```

Pi colour values run from 0 to 1; picozero colour values run from 0 to 255. These examples make magenta.

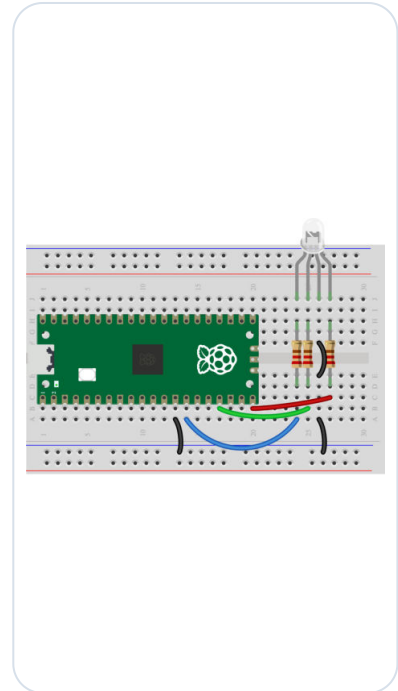
Wiring

RASPBERRY PI 5



R → **GPIO17**, G → **GPIO27**, B → **GPIO22**;
common → **GND**

PICO



R → **GP15**, G → **GP13**, B → **GP10**;
common → **GND**

NeoPixels

DO

OUTPUT

LIGHT

What does it do?

A NeoPixel is a full-colour LED with a tiny controller built into it.

One data wire carries colour instructions, and each pixel passes the remaining instructions onwards.

The 3.3 V-compatible JST breakout shown here can connect directly to a Pi or Pico.

Uses



Gaming PC lighting

DETECT

Read lighting selection



DECIDE

Which colour should each pixel be?



DO

Light individual pixels



Disco shoelace meter

DETECT

Read shoe movement



DECIDE

Which light pattern should run?



DO

Animate the pixels



Pixel
RGB LED and controller

IN
Data from the board

OUT
Data to another pixel

VDD and GND
3.3 V power

Code

● Raspberry Pi

```
import board
import neopixel

pixels = neopixel.NeoPixel(board.D18, 1)

pixels[0] = (255, 0, 80)
```

● Raspberry Pi Pico

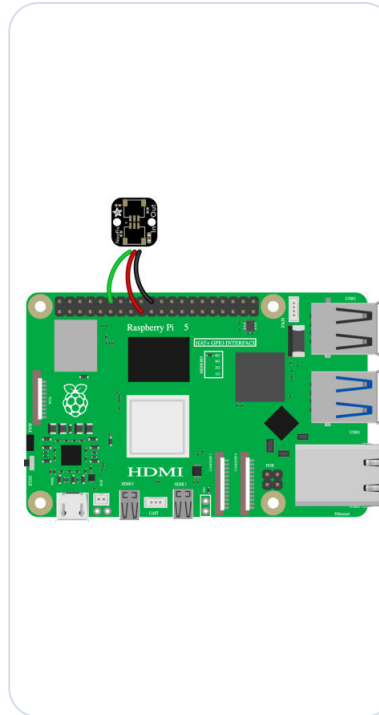
```
from machine import Pin
from neopixel import NeoPixel

pixels = NeoPixel(Pin(28), 1)
pixels[0] = (255, 0, 80)
pixels.write()
```

This wiring is for the pictured 3.3 V-compatible breakout. Generic 5 V strips usually need a separate supply and a logic-level shifter.

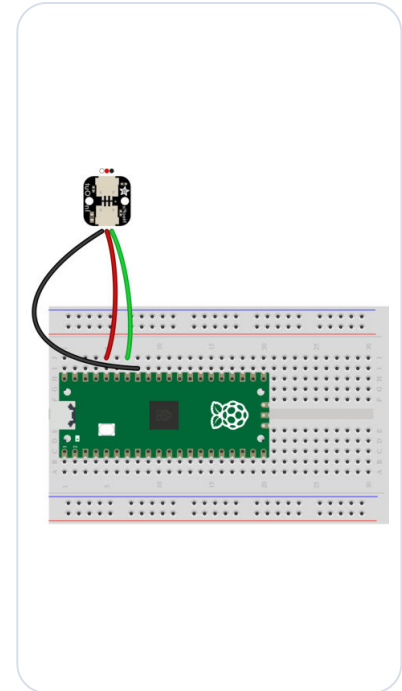
Wiring

RASPBERRY PI 5



GPIO18 → IN, **3V3** → VDD, **GND** → GND

PICO



GP28 → IN, **3V3 OUT** → VDD, **GND** → GND

Buzzer

DO

SOUND

What does it do?

A buzzer makes a sound the moment your code switches it on.

An active buzzer has its tone built in, so it beeps at one fixed pitch. You switch it on and off rather than choosing notes.

Your projects can use it for an alarm, a warning, a countdown, or a fanfare when somebody wins.

Uses



Smoke alarm

DETECT

Measure smoke level



DECIDE

Is smoke above the limit?



DO

Sound the buzzer



Buzz wire game

DETECT

Loop touches wire



DECIDE

Is this the third strike?



DO

Sound the failure buzzer



+ leg

Longer, to GPIO

- leg

Shorter, to GND

Vent

Sound comes out

Code

● Raspberry Pi

```
from gpiozero import Buzzer  
  
buzzer = Buzzer(12)  
  
buzzer.on()
```

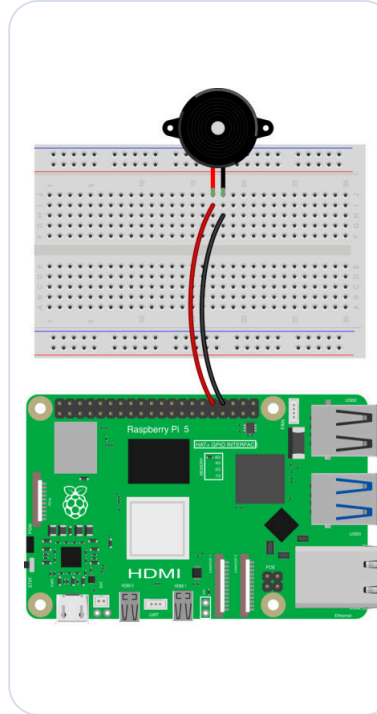
● Raspberry Pi Pico

```
from picozero import Buzzer  
  
buzzer = Buzzer(16)  
  
buzzer.on()
```

Also `buzzer.off()`, `buzzer.toggle()` and `buzzer.beep()`. Give `beep()` an `on_time` and an `off_time` to set the rhythm.

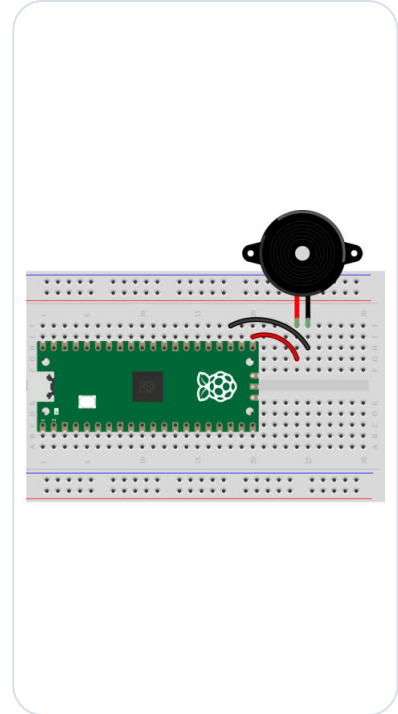
Wiring

RASPBERRY PI 5



GPIO 12 → + buzzer - → GND

PICO



GP16 → + buzzer - → GND

Speaker

DO

OUTPUT

SOUND

What does it do?

A small passive speaker turns a changing electrical signal into sound.

Current through its coil moves a cone backwards and forwards, pushing the air.

The 220 ohm resistor shown here protects the GPIO; it also keeps this simple circuit fairly quiet.

Uses



Door chime

DETECT
Button and
chime setting



DECIDE
Which chime
should play?



DO
Play the notes



Tiny fanfare machine

DETECT
Count
achievement-
button presses



DECIDE
Has the
target been
reached?



DO
Play
triumphant
notes



Cone
Pushes the air

Coil and magnet
Create
movement

Terminals
Connect
through 220
ohms

Code

● Raspberry Pi

```
from gpiozero import TonalBuzzer  
speaker = TonalBuzzer(17)  
speaker.play("E4")
```

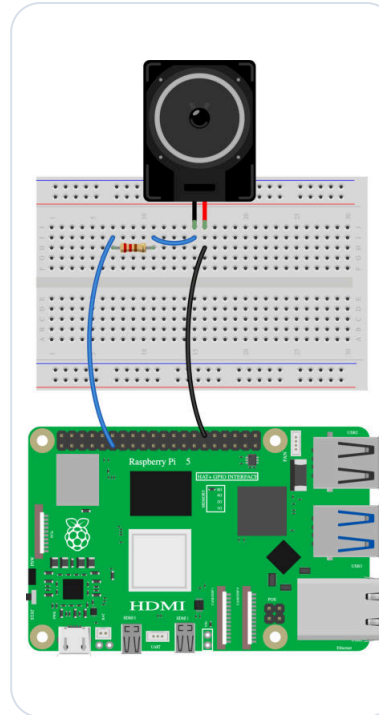
● Raspberry Pi Pico

```
from picozero import Speaker  
speaker = Speaker(16)  
speaker.play("E4", duration=1)
```

For a louder speaker, use a transistor or amplifier rather than removing the protective resistor.

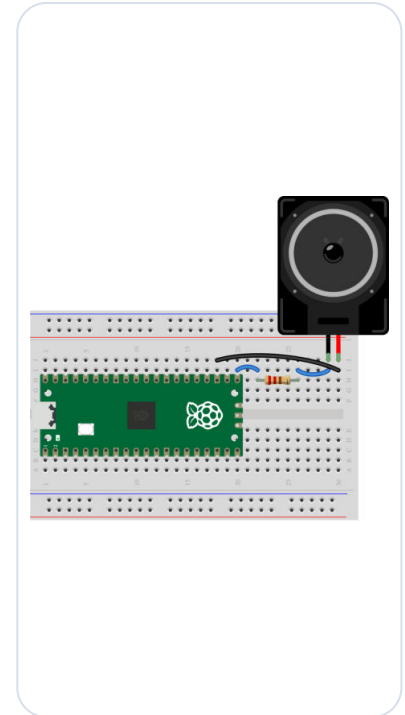
Wiring

RASPBERRY PI 5



GPIO17 → 220 ohms → speaker → **GND**

PICO



GP16 → 220 ohms → speaker → **GND**

Motor driver

DO

OUTPUT

MOVEMENT

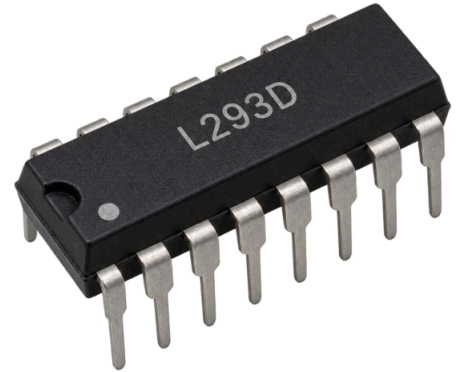
What does it do?

A motor driver lets a low-power GPIO control a motor that needs more current.

The L293D contains H-bridges that can send current through a motor in either direction.

Two input pins choose forwards, backwards or stop, while the chip handles the motor current.

Uses

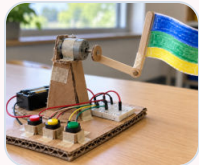


Electric car window

DETECT
Read window
switch

DECIDE
Can it move
safely?

DO
Drive
requested
direction



Tiny flag waver

DETECT
Count score-
button presses

DECIDE
Has the
target been
reached?

DO
Wave the flag

Notch
Identifies
pin 1

**IN1 and
IN2**
Direction
controls

**OUT1
and
OUT2**
Connect
to the
motor

Enable
Held high
in this
circuit

Code

● Raspberry Pi

```
from gpiozero import Motor
motor = Motor(forward=17, backward=18)
motor.forward()
```

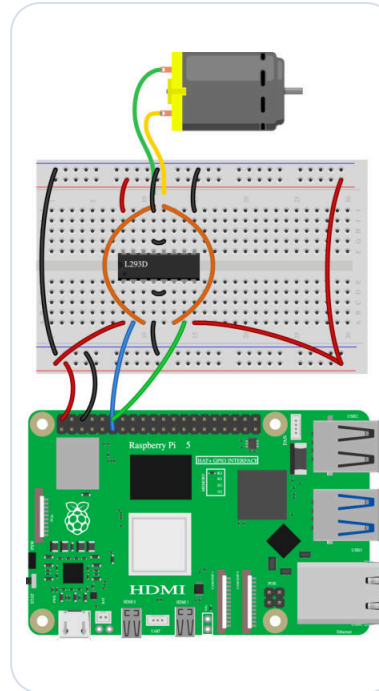
● Raspberry Pi Pico

```
from picozero import Motor
motor = Motor(14, 15)
motor.forward()
```

This card powers one small 5 V motor from the board's 5 V rail. Use a separate motor supply for larger motors, with the grounds joined.

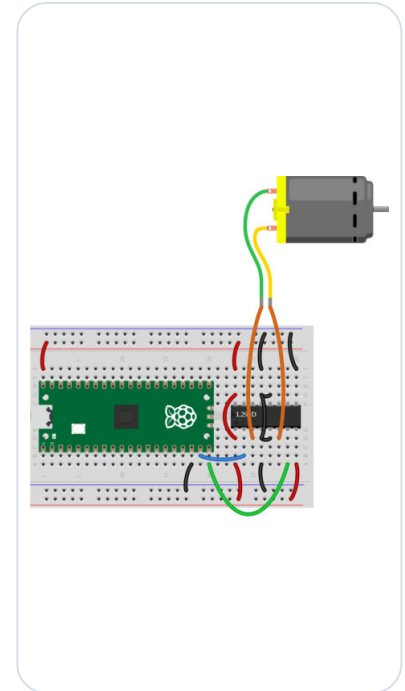
Wiring

RASPBERRY PI 5



GPIO17 → IN1, **GPIO18** → IN2; OUT1 and OUT2 → motor

PICO



GP14 → IN1, **GP15** → IN2; OUT1 and OUT2 → motor

Stepper motor

DO

OUTPUT

MOVEMENT

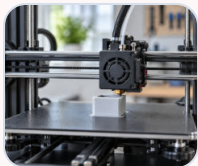
What does it do?

A stepper motor turns in small, repeatable steps instead of spinning freely.

Energising its coils in sequence moves the shaft one step at a time and holds its position.

The ULN2003 board supplies the coil current for the common 28BYJ-48 motor shown here.

Uses



3D printer positioning

DETECT
Read next print position

DECIDE
How many steps are needed?

DO
Move the print head

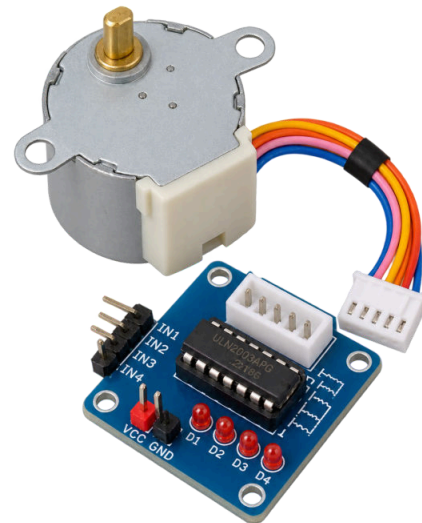


Snack lift

DETECT
Snack request arrives

DECIDE
Which floor needs it?

DO
Step lift into place



Output shaft

Moves in measured steps

Five-wire plug

Connects to the driver

IN1 to IN4

Coil sequence

5V and GND

Driver power

Code

● Raspberry Pi

```
from gpiozero import OutputDevice
from time import sleep

coils = [OutputDevice(pin) for pin in (22, 27, 17, 4)]
sequence = ((1, 0, 0, 0), (0, 1, 0, 0),
            (0, 0, 1, 0), (0, 0, 0, 1))

for pattern in sequence:
    for coil, value in zip(coils, pattern):
        coil.value = value
    sleep(0.002)
```

● Raspberry Pi Pico

```
from picozero import Stepper

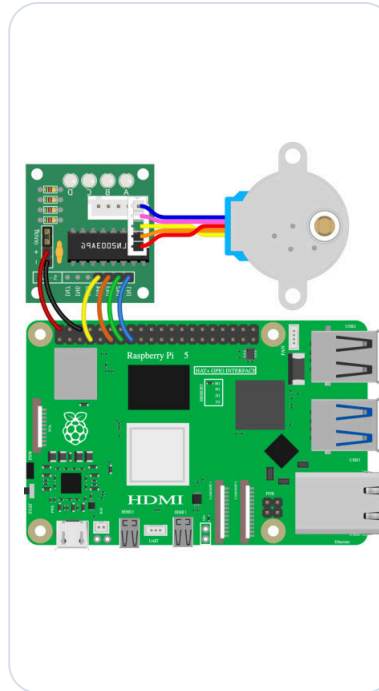
motor = Stepper((16, 17, 18, 19))

motor.step(100)
```

Repeat the Pi sequence to keep moving.
Reverse either board's sequence to turn the other way.

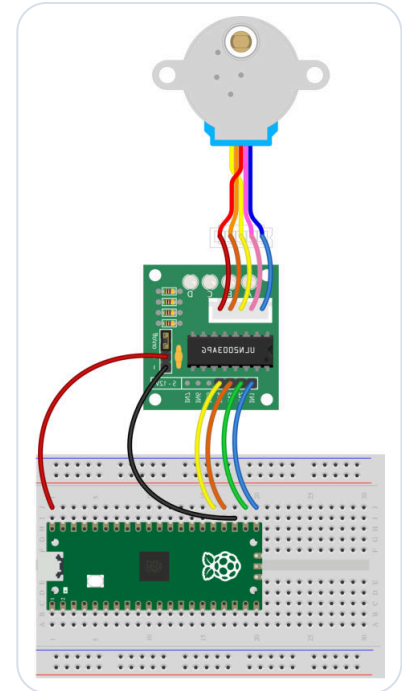
Wiring

RASPBERRY PI 5



IN1-IN4 → GPIO **22, 27, 17, 4**; driver →
5V and **GND**

PICO



IN1-IN4 → GP **16, 17, 18, 19**; driver →
VBUS and **GND**